Vilhes Samy-Melwan

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Summary

Pursuing a PhD at LITIS Lab, INSA Rouen Normandie (France), researching anomaly detection in time series and developing foundation models for zero-shot time-series forecasting. Strong mathematical background: B.Sc. and M.Sc. in Mathematics and Artificial Intelligence from Université Paris-Saclay.

Work Experience

PhD in A.I. at LITIS Lab, INSA Rouen Normandie November 2024 - present

Research on deep-learning methods for time series: implementation and benchmarking of state-of-the-art approaches for anomaly detection and forecasting; development of foundation models for zero-shot forecasting.

Teaching: Assistant Professor in Signal Processing and Data Analysis for 3rd-year engineering students.

A.I. Researcher Intern at Thales April 2024 - September 2024

Researched the Neural Tangent Kernel as a training-free metric for neural architecture search; evaluated and compared search strategies for efficient architecture exploration.

A.I. Engineer Intern at AZAP March 2023 - August 2023

Implemented and benchmarked machine-learning models (best: gradient boosting) to forecast sales of promoted products; focused on uncertainty analysis to improve forecast reliability.

Education

2024 - present	PhD (Subject) at **LITIS Lab, INSA Rouen Normandie**
2022 - 2024	Master's Degree in Mathematics and Artificial Intelligence at **Université Paris-Saclay**
2019 - 2022	Bachelor's Degree in Mathematics at **Université Paris-Saclay**

Publications

Vilhes, Samy-Melwan et al. (2025). "PatchTrAD: A Patch-Based Transformer focusing on Reconstruction Error for Time Series Anomaly Detection". In: *EUSIPCO 2025*. URL: `https://arxiv.org/abs/2504.08827`.

Vilhes, Samy-Melwan (2026). "Understanding Neural Tangent Kernel : Key Theories and Experimental Insights". URL: `https://normandie-univ.hal.science/hal-04784111`.

Vilhes, Samy-Melwan et al. (2026). "Do Normalization Choice Matter for Causal Large Time-Series Models?" In: *1st ICLR Workshop on Time Series in the Age of Large Models*. URL: `https://openreview.net/forum?id=lMNWBnFHxt`.

Skills

High-performance computing	experience with HPC infrastructure (e.g., Jean Zay, CNRS; CRIANN, Normandie).
Handling large-scale models	handling models up to 1B parameters.
Mathematics	huge understanding of mathematical foundations behind deep learning.
PyTorch	extensive experience implementing diverse deep-learning architectures.

Summer Schools & Conferences

EUSIPCO 2025	Rank B international conference where I presented *PatchTrAD*. Slides
Hi! Paris Summer School 2025	Hosted at École Polytechnique, Paris, with a poster session. Poster